

Delta Method and Method of Moments

Clément Lalanne

Notation for this lecture

Parameter and observations. The parameter is $\theta \in \Theta \subset \mathbb{R}^p$, the observations $X_1, X_2, \dots$ are i.i.d. copies of a random vector X of $\mathbb{R}^d$, and $\hat{\theta}_n$ denotes an estimator built from $X_1, \dots, X_n$.

Derivatives. For $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^m$ differentiable at θ , we denote by $J\phi(\theta)$ its $m \times p$ Jacobian matrix at θ . When $m = 1$, $\nabla\phi(\theta)$ denotes the gradient column vector and $H\phi(\theta)$ the Hessian.

Model. A statistical model is $\{\mathcal{L}_\theta; \theta \in \Theta\}$, each $\mathcal{L}_\theta$ a law on $\mathbb{R}^d$. We write $\mathbb{E}_{X \sim \mathcal{L}_\theta}$, abbreviated $\mathbb{E}_\theta$, for expectations computed as if the parameter were θ , which is the only abbreviation of this lecture.

Sample quantities. We denote $\bar{X}_n = \frac{1}{n} \sum_{i \leq n} X_i$, and any sample dependent object carries its index (e.g. $\hat{\theta}_n$, $\hat{\mu}_{2,n}$, e_n).

Part I. The delta method

The question

Setup. Given a sequence of estimators $(\hat{\theta}_n)$ of a parameter θ and a mapping ϕ , it is natural to estimate $\phi(\theta)$ by $\phi(\hat{\theta}_n)$.

Question. We ask whether the asymptotic properties of $\hat{\theta}_n - \theta$ are inherited by $\phi(\hat{\theta}_n) - \phi(\theta)$.

Consistency. Here the answer is immediate. If $\hat{\theta}_n \xrightarrow{\mathbb{P}} \theta$ and ϕ is continuous at θ , then $\phi(\hat{\theta}_n) \xrightarrow{\mathbb{P}} \phi(\theta)$ by continuous mapping, since the continuity set contains the constant θ .

Rate and limit law. Let us now suppose that

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma) .$$

In contrast, nothing above says at what rate $\phi(\hat{\theta}_n) - \phi(\theta)$ goes to zero, nor what its limit law is.

The linear case

If ϕ is linear, given by an $m \times p$ matrix M , then $\phi(\hat{\theta}_n) - \phi(\theta) = M(\hat{\theta}_n - \theta)$, and we have

$$\begin{aligned}\sqrt{n}(M\hat{\theta}_n - M\theta) &= M\sqrt{n}(\hat{\theta}_n - \theta) \\ &\xrightarrow{d} M\mathcal{N}(0, \Sigma) \quad \text{by continuous mapping with } x \mapsto Mx \\ &= \mathcal{N}(0, M\Sigma M^\top) .\end{aligned}$$

The last equality follows from the characteristic function computation of lecture 2, since for $Z \sim \mathcal{N}(0, \Sigma)$ and $t \in \mathbb{R}^m$, we have $\mathbb{E}_Z[e^{it^\top MZ}] = \mathbb{E}_Z[e^{i(M^\top t)^\top Z}] = \exp(-\frac{1}{2}t^\top M\Sigma M^\top t)$.

The intuition of the delta method. A function differentiable at θ is, near θ , roughly its linearisation. Hence, since $\hat{\theta}_n$ concentrates at θ , only the behaviour of ϕ at θ should matter, and $J\phi(\theta)$ should play the role of M .

The theorem

Theorem (delta method). Let $\theta \in \mathbb{R}^p$ and let $\phi : \mathbb{R}^p \rightarrow \mathbb{R}^m$ be differentiable at θ . Let $(\hat{\theta}_n)$ be a sequence of random vectors of $\mathbb{R}^p$, X a random vector of $\mathbb{R}^p$, and (r_n) real numbers with $r_n \rightarrow \infty$. If $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$, then

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} (J\phi(\theta))X ,$$

and, more precisely,

$$r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) - r_n(J\phi(\theta))(\hat{\theta}_n - \theta) \xrightarrow{\mathbb{P}} 0 .$$

Remark. The second statement is the stronger one, since it says that the linearisation is exact up to a term negligible in probability at the scale r_n . Hence, the proof establishes the second statement and deduces the first from it.

Proof of the delta method

We denote by $R(h) = \phi(\theta + h) - \phi(\theta) - (J\phi(\theta))h$ the remainder of the first order expansion at θ , so that $R(0) = 0$ and $\|R(h)\| = o(\|h\|)$ as $h \rightarrow 0$, which is the definition of differentiability at θ .

Hence, we have

$$\begin{aligned} & r_n \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \\ &= (J\phi(\theta))r_n(\hat{\theta}_n - \theta) + r_n R(\hat{\theta}_n - \theta) \\ &= (J\phi(\theta))r_n(\hat{\theta}_n - \theta) + o_{\mathbb{P}} \left(\|r_n(\hat{\theta}_n - \theta)\| \right) && \text{by the transfer theorem with } q = 1 \\ &= (J\phi(\theta))r_n(\hat{\theta}_n - \theta) + o_{\mathbb{P}}(1) && \text{by Prohorov} \\ &\xrightarrow{d} (J\phi(\theta))X . && \text{by Slutsky and continuous mapping} \end{aligned}$$

Reading the hypotheses

Differentiability is needed at one point only. We assume nothing about ϕ away from θ , not even continuity, which is possible since $\hat{\theta}_n$ concentrates at θ .

The limit X need not be gaussian. The theorem holds for an arbitrary limit in law. However, in every application below, X is gaussian since it comes from the central limit theorem, and then $(J\phi(\theta))X \sim \mathcal{N}(0, J\phi(\theta)\Sigma(J\phi(\theta))^\top)$.

The rate is unchanged. Since the same r_n appears on both sides, $\phi(\hat{\theta}_n)$ converges at the rate of $\hat{\theta}_n$, with a different constant.

The case of a vanishing Jacobian. If $J\phi(\theta) = 0$, the limit is the constant 0 and the theorem, while true, only says that the rate r_n is too slow. We give the right rate in Part III.

Exercise, the inverse of a mean

Exercise. Let $X_1, X_2, \dots$ be i.i.d. of exponential law of parameter $\lambda > 0$, i.e. of density $x \mapsto \lambda e^{-\lambda x}$ on $(0, \infty)$, so that $\mathbb{E}_X[X] = 1/\lambda$ and $\text{Var}_X(X) = 1/\lambda^2$. Find the limit law of $\sqrt{n}(1/\bar{X}_n - \lambda)$.

Solution. By the central limit theorem, $\sqrt{n}(\bar{X}_n - 1/\lambda) \xrightarrow{d} \mathcal{N}(0, 1/\lambda^2)$. The map $\phi(x) = 1/x$ is differentiable at $1/\lambda$ with $\phi'(1/\lambda) = -\lambda^2$, so that we have

$$\begin{aligned} \sqrt{n} \left(\frac{1}{\bar{X}_n} - \lambda \right) &= \sqrt{n} (\phi(\bar{X}_n) - \phi(1/\lambda)) \\ &\xrightarrow{d} \mathcal{N}(0, \phi'(1/\lambda)^2/\lambda^2) \quad \text{by the delta method with } r_n = \sqrt{n} \\ &= \mathcal{N}(0, \lambda^2) . \end{aligned}$$

Remark. The estimator is well defined since the X_i are positive almost surely. Its limiting variance depends on the unknown λ , which is the situation that variance stabilisation addresses in Part III.

Part II. Estimating a variance

The problem

Let $X_1, X_2, \dots$ be i.i.d. copies of a real X with $\mathbb{E}_X[|X|^4] < \infty$. We denote $\mu_1 = \mathbb{E}_X[X]$ and, for $\ell \in \{2, 3, 4\}$, the centered moments $\mu_\ell = \mathbb{E}_X[(X - \mu_1)^\ell]$.

We estimate the mean by $\hat{\mu}_{1,n} = \bar{X}_n$ and the variance μ_2 by the empirical variance

$$\hat{\mu}_{2,n} = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2 .$$

The difficulty. Since $\hat{\mu}_{1,n}$ is an average of i.i.d. variables, the central limit theorem applies to it directly. In contrast, $\hat{\mu}_{2,n}$ is not such an average, since its summands all involve $\bar{X}_n$, hence all the data, and they are identically distributed but not independent.

An identity for the empirical variance

We have

$$\begin{aligned}\hat{\mu}_{2,n} &= \frac{1}{n} \sum_{i \leq n} X_i^2 - \left(\frac{1}{n} \sum_{i \leq n} X_i \right)^2 && \text{by expanding the square} \\ &= \frac{1}{n} \sum_{i \leq n} (X_i - \mu_1)^2 - \left(\frac{1}{n} \sum_{i \leq n} (X_i - \mu_1) \right)^2 && \text{by invariance of a variance under translation}\end{aligned}$$

The right side is thus a fixed function of two averages of i.i.d. variables. Let us set $\phi(x, y) = y - x^2$ and define the vectors of $\mathbb{R}^2$

$$Y_i = \begin{pmatrix} X_i - \mu_1 \\ (X_i - \mu_1)^2 \end{pmatrix},$$

so that $\hat{\mu}_{2,n} = \phi(\bar{Y}_n)$ exactly, and $\mu_2 = \phi(0, \mu_2)$.

The central limit theorem for the pair

Since $\mathbb{E}_X[(X - \mu_1)] = 0$ and $\mathbb{E}_X[(X - \mu_1)^2] = \mu_2$, the vector Y_1 has mean $(0, \mu_2)^\top$ and covariance $\Sigma_Y := \text{Cov}_X(Y_1)$, given by

$$\Sigma_Y = \begin{pmatrix} \mu_2 & \mu_3 \\ \mu_3 & \mu_4 - \mu_2^2 \end{pmatrix},$$

which is finite since $\mathbb{E}_X[|X|^4] < \infty$. The entries are $\text{Var}_X(X - \mu_1) = \mu_2$, then $\mathbb{E}_X[(X - \mu_1)^3] = 0 \cdot \mu_2 = \mu_3$, and $\text{Var}_X((X - \mu_1)^2) = \mu_4 - \mu_2^2$.

Since the Y_i are i.i.d. and square integrable, the central limit theorem in $\mathbb{R}^2$ yields

$$\sqrt{n} (\bar{Y}_n - (0, \mu_2)^\top) \xrightarrow{d} \mathcal{N}(0, \Sigma_Y).$$

Applying the delta method

The map $\phi(x, y) = y - x^2$ is differentiable with $J\phi(x, y) = (-2x, 1)$, so that at the evaluation point, $J\phi(0, \mu_2) = (0, 1)$. Hence, we have

$$\begin{aligned}\sqrt{n}(\hat{\mu}_{2,n} - \mu_2) &= \sqrt{n}(\phi(\bar{Y}_n) - \phi(0, \mu_2)) \\ &\xrightarrow{d} \mathcal{N}(0, (0, 1)\Sigma_Y(0, 1)^\top) \quad , \quad \text{by the delta method with } r_n = \sqrt{n}\end{aligned}$$

and the quadratic form picks out the lower right entry:

$$\sqrt{n}(\hat{\mu}_{2,n} - \mu_2) \xrightarrow{d} \mathcal{N}(0, \mu_4 - \mu_2^2) \quad .$$

The first coordinate of Y has disappeared since $J\phi$ vanishes there. In other words, the fluctuation of $\bar{X}_n$ contributes nothing at first order to the fluctuation of the empirical variance.

Remarks on the result

The limiting variance is $\text{Var}_X((X - \mu_1)^2)$. It is zero if and only if $(X - \mu_1)^2$ is almost surely constant, i.e. when X takes only the two values $\mu_1 \pm \sqrt{\mu_2}$, and for every other law, the limit is nondegenerate.

In the gaussian case, $\mu_4 = 3\mu_2^2$, so the limit is $\mathcal{N}(0, 2\mu_2^2)$, and the relative precision $\sqrt{n}(\hat{\mu}_{2,n} - \mu_2)/\mu_2$ has limiting variance 2 whatever the scale.

The bias correction. With $s_n^2 = \frac{n}{n-1}\hat{\mu}_{2,n}$ the unbiased version, we have $\sqrt{n}(s_n^2 - \hat{\mu}_{2,n}) = \frac{\sqrt{n}}{n-1}\hat{\mu}_{2,n} \xrightarrow{\mathbb{P}} 0$, so that by asymptotic equivalence, s_n^2 has the same limit law. Hence, unbiasedness at fixed n has no asymptotic content here.

Exercise, the empirical standard deviation

Exercise. Under the assumptions of this part, and if $\mu_2 > 0$, find the limit law of $\sqrt{n}(\sqrt{\hat{\mu}_{2,n}} - \sqrt{\mu_2})$. Specialise it to the gaussian case.

Solution. The map $\phi(t) = \sqrt{t}$ is differentiable at $\mu_2 > 0$ with $\phi'(\mu_2) = 1/(2\sqrt{\mu_2})$. Hence, starting from the limit law of the empirical variance, we have

$$\begin{aligned}\sqrt{n} \left(\sqrt{\hat{\mu}_{2,n}} - \sqrt{\mu_2} \right) &= \sqrt{n} (\phi(\hat{\mu}_{2,n}) - \phi(\mu_2)) \\ &\xrightarrow{d} \mathcal{N} \left(0, \frac{\mu_4 - \mu_2^2}{4\mu_2} \right) . \quad \text{by the delta method with } r_n = \sqrt{n}\end{aligned}$$

In the gaussian case, $\mu_4 = 3\mu_2^2$, and the limit is $\mathcal{N}(0, \mu_2/2)$.

Remark. We applied the delta method twice, which is legitimate since its only input is a limit in law. Equivalently, we can apply it once to $(x, y) \mapsto \sqrt{y - x^2}$ at $(0, \mu_2)$, whose Jacobian there is $(0, 1/(2\sqrt{\mu_2}))$ by the chain rule.

Part III. Two refinements

The case of a vanishing Jacobian

If $J\phi(\theta) = 0$, the delta method yields $r_n(\phi(\hat{\theta}_n) - \phi(\theta)) \xrightarrow{d} 0$, a true but empty statement which only says that the rate r_n is too slow. We now show that the correct rate is r_n^2 .

Theorem (second order delta method). Let $\phi : \mathbb{R}^p \rightarrow \mathbb{R}$ be twice differentiable at θ with $\nabla\phi(\theta) = 0$, and let $r_n(\hat{\theta}_n - \theta) \xrightarrow{d} X$ with $r_n \rightarrow \infty$. Then

$$r_n^2 \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} \frac{1}{2} X^\top (H\phi(\theta)) X .$$

The limit is no longer gaussian. Even for X gaussian, a quadratic form in a gaussian vector is a weighted sum of independent χ_1^2 variables, with weights the eigenvalues of $\frac{1}{2}\Sigma^{1/2}(H\phi(\theta))\Sigma^{1/2}$.

Proof of the second order version

Using $\nabla\phi(\theta) = 0$, we denote by $R(h) = \phi(\theta + h) - \phi(\theta) - \frac{1}{2}h^\top (H\phi(\theta))h$ the remainder of the second order expansion at θ , so that $R(0) = 0$ and $|R(h)| = o(\|h\|^2)$ as $h \rightarrow 0$, which is the Taylor expansion at order two of a function twice differentiable at θ .

Hence, denoting $Z_n = r_n(\hat{\theta}_n - \theta)$, we have

$$\begin{aligned} & r_n^2 \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \\ &= \frac{1}{2} Z_n^\top (H\phi(\theta)) Z_n + r_n^2 R(\hat{\theta}_n - \theta) \\ &= \frac{1}{2} Z_n^\top (H\phi(\theta)) Z_n + o_{\mathbb{P}}(\|Z_n\|^2) && \text{by the transfer theorem with } q = 2 \\ &= \frac{1}{2} Z_n^\top (H\phi(\theta)) Z_n + o_{\mathbb{P}}(1) && \text{by Prohorov} \\ &\xrightarrow{d} \frac{1}{2} X^\top (H\phi(\theta)) X . && \text{by Slutsky and continuous mapping} \end{aligned}$$

Example, a proportion at one half

Let $X_1, \dots, X_n$ be i.i.d. $\text{Ber}(\theta)$ and $\phi(\theta) = \theta(1 - \theta)$, the variance of a single observation. Then $\phi'(\theta) = 1 - 2\theta$, which vanishes only at $\theta = 1/2$.

Away from one half, the delta method yields $\sqrt{n}(\bar{X}_n(1 - \bar{X}_n) - \theta(1 - \theta)) \xrightarrow{d} \mathcal{N}(0, (1 - 2\theta)^2\theta(1 - \theta))$.

At $\theta = 1/2$, the first order limit is degenerate, and applying the second order version with $\phi'' = -2$ and $\sqrt{n}(\bar{X}_n - \frac{1}{2}) \xrightarrow{d} \mathcal{N}(0, \frac{1}{4})$, we obtain, with $Z \sim \mathcal{N}(0, \frac{1}{4})$,

$$\begin{aligned} n(\bar{X}_n(1 - \bar{X}_n) - 1/4) &\xrightarrow{d} -Z^2 \\ &\stackrel{d}{=} -\frac{1}{4}\chi_1^2. \end{aligned}$$

The rate improves from $n^{-1/2}$ to n^{-1} and the limit is one sided, as expected since $\bar{X}_n(1 - \bar{X}_n) \leq 1/4$ always, so that the error can only have one sign.

Exercise, a cosine at zero

Exercise. Let $X_1, X_2, \dots$ be i.i.d. real with $\mathbb{E}_X[X] = 0$ and $\text{Var}_X(X) = \sigma^2 \in (0, \infty)$. Find the limit law of $\sqrt{n}(\cos \bar{X}_n - 1)$, then the correct rate and the corresponding limit law.

Solution. By the central limit theorem, $\sqrt{n}\bar{X}_n \xrightarrow{d} \mathcal{N}(0, \sigma^2)$. Since $\cos'(0) = -\sin 0 = 0$, the delta method yields $\sqrt{n}(\cos \bar{X}_n - 1) \xrightarrow{d} 0$, so that the rate $n^{-1/2}$ is too slow.

Since $\cos''(0) = -1$, the second order version with $r_n = \sqrt{n}$ yields, with $Z \sim \mathcal{N}(0, \sigma^2)$,

$$\begin{aligned} n(\cos \bar{X}_n - 1) &\xrightarrow{d} -\frac{1}{2}Z^2 \\ &\stackrel{d}{=} -\frac{\sigma^2}{2}\chi_1^2. \end{aligned}$$

The limit is nonpositive, as it should be since $\cos \leq 1$.

Variance stabilisation

Let us take $p = m = 1$ and suppose that $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2(\theta))$, with a limiting variance depending on the unknown θ . For ϕ differentiable, the delta method yields

$$\sqrt{n} \left(\phi(\hat{\theta}_n) - \phi(\theta) \right) \xrightarrow{d} \mathcal{N} \left(0, \phi'(\theta)^2 \sigma^2(\theta) \right) .$$

The recipe. We choose ϕ with $\phi'(\theta) = 1/\sigma(\theta)$ for every θ , i.e. a primitive of $1/\sigma$. Then the limit is $\mathcal{N}(0, 1)$, the same for every θ , so that the unknown no longer appears in the limit law.

Advantages. Since no plug-in estimate of the variance is needed, the studentisation step of lecture 2 disappears. Furthermore, the normal approximation is usually accurate at smaller n , since the transformed statistic is closer to symmetric.

The arcsine transformation

For $X_1, \dots, X_n$ i.i.d. $\text{Ber}(\theta)$ with $\theta \in (0, 1)$, $\sigma^2(\theta) = \theta(1 - \theta)$, so that the recipe asks for a primitive of $1/\sqrt{\theta(1 - \theta)}$, namely

$$\phi(\theta) = 2 \arcsin \sqrt{\theta} ,$$

since $\phi'(\theta) = \frac{2}{\sqrt{1-\theta}} \cdot \frac{1}{2\sqrt{\theta}} = 1/\sqrt{\theta(1 - \theta)}$, and hence $\sqrt{n} \left(2 \arcsin \sqrt{\bar{X}_n} - 2 \arcsin \sqrt{\theta} \right) \xrightarrow{d} \mathcal{N}(0, 1)$.

The interval. With r_α the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, inverting ϕ , which is increasing from $[0, 1]$ onto $[0, \pi]$, yields an asymptotic confidence interval of level $1 - \alpha$ for θ , where the angles are clipped to $[0, \pi/2]$:

$$\left[\sin^2 \left(\max \left(0, \arcsin \sqrt{\bar{X}_n} - r_\alpha / (2\sqrt{n}) \right) \right), \sin^2 \left(\min \left(\pi/2, \arcsin \sqrt{\bar{X}_n} + r_\alpha / (2\sqrt{n}) \right) \right) \right] .$$

Other classical cases. Poisson with $\sigma^2(\lambda) = \lambda$ leads to $\phi(\lambda) = 2\sqrt{\lambda}$, and the correlation coefficient to the Fisher transformation $\phi(\rho) = \text{arctanh}(\rho)$.

Exercise, stabilising an exponential rate

Exercise. For $\hat{\lambda}_n = 1/\bar{X}_n$, the exercise on the inverse of a mean yields $\sqrt{n}(\hat{\lambda}_n - \lambda) \xrightarrow{d} \mathcal{N}(0, \lambda^2)$. Find a variance stabilising transformation, and deduce an asymptotic confidence interval of level $1 - \alpha$ for λ .

Solution. Here $\sigma(\lambda) = \lambda$, and a primitive of $1/\lambda$ on $(0, \infty)$ is $\phi = \log$. Since $\phi'(\lambda)^2 \sigma^2(\lambda) = 1$, the delta method yields

$$\sqrt{n} \left(\log \hat{\lambda}_n - \log \lambda \right) \xrightarrow{d} \mathcal{N}(0, 1) .$$

With r_α the quantile of order $1 - \alpha/2$ of $\mathcal{N}(0, 1)$, the probability that $|\log \hat{\lambda}_n - \log \lambda| \leq r_\alpha / \sqrt{n}$ thus tends to $1 - \alpha$ by portmanteau, and exponentiating yields the interval

$$\left[\hat{\lambda}_n e^{-r_\alpha / \sqrt{n}}, \hat{\lambda}_n e^{r_\alpha / \sqrt{n}} \right] .$$

It lies in $(0, \infty)$ for every n , in contrast with the plug-in interval $\hat{\lambda}_n(1 \pm r_\alpha / \sqrt{n})$, whose lower end is negative when $n < r_\alpha^2$.

Part IV. Statistical models

The model

Definition. A parametric statistical model is a family $\{\mathcal{L}_\theta; \theta \in \Theta\}$ with $\Theta \subset \mathbb{R}^p$, each $\mathcal{L}_\theta$ a law on $\mathbb{R}^d$. In other words, it is a set of candidate laws for X_1 .

Well specified. We assume throughout that the model contains the truth, i.e. that there is θ_0 in the interior of Θ such that $X_1, \dots, X_n$ are i.i.d. of law $\mathcal{L}_{\theta_0}$, and our goal is to estimate θ_0 .

Notation. $\mathbb{E}_\theta, \mathbb{P}_\theta, \text{Cov}_\theta$ denote the expectation, probability and covariance computed as if the parameter were θ . For instance, with $d = 1, \Theta = (0, \infty)$ and $\mathcal{L}_\theta = \mathcal{N}(0, \theta)$, we have

$$\begin{aligned}\mathbb{E}_3[X_1^2] &= \int_{\mathbb{R}} x^2 d\mathcal{L}_3(x) \\ &= \mathbb{E}_{Z \sim \mathcal{N}(0,3)}[Z^2] \\ &= 3 .\end{aligned}$$

We keep writing $\mathbb{E}_{\theta_0} = \mathbb{E}_X$, since $\mathcal{L}_{\theta_0}$ really is the law of the data.

Identifiability

A method that matches p features of the law can only work if those features separate the parameters. Given $f_1, \dots, f_p : \mathbb{R}^d \rightarrow \mathbb{R}$, we denote by $e : \Theta \rightarrow \mathbb{R}^p$ the map

$$e(\theta) = \begin{pmatrix} \mathbb{E}_\theta[f_1(X_1)] \\ \vdots \\ \mathbb{E}_\theta[f_p(X_1)] \end{pmatrix}.$$

Identifiability from the moments is the requirement that $e(\theta) \neq e(\theta_0)$ for every $\theta \neq \theta_0$. Without it, two parameters produce identical moment equations, and no sample size separates them.

The theorem below assumes instead that e is continuously differentiable near θ_0 with $Je(\theta_0)$ invertible. By the inverse function theorem, this only forces local injectivity, which does not imply identifiability, and makes e locally invertible, which is what the estimator relies on.

Number of moment functions. There are as many moment functions as parameters (p of each), since $Je(\theta_0)$ is $p \times p$ and has to be invertible. With fewer, the system is underdetermined, whereas with more, it is generally incompatible.

Exercise, a gaussian scale model

Exercise. Let $\Theta = \mathbb{R} \setminus \{0\}$ and $\mathcal{L}_\theta = \mathcal{N}(0, \theta^2)$, with $p = 1$. Compute e , and decide whether θ_0 is identifiable from the moments and whether $Je(\theta_0)$ is invertible, first with $f_1(x) = x$, then with $f_1(x) = x^2$.

With $f_1(x) = x$, we have $e(\theta) = \mathbb{E}_\theta[X_1] = 0$ for every θ . Hence, e is constant, so that θ_0 is not identifiable and $Je(\theta_0) = 0$ is not invertible.

With $f_1(x) = x^2$, we have $e(\theta) = \mathbb{E}_\theta[X_1^2] = \theta^2$, so that $Je(\theta_0) = 2\theta_0 \neq 0$ is invertible. However, $e(-\theta_0) = e(\theta_0)$, so that θ_0 is not identifiable, which is an instance of local injectivity without identifiability.

The cause is the model. Since $\mathcal{L}_{-\theta} = \mathcal{L}_\theta$, no method separates θ_0 from $-\theta_0$. With $\Theta = (0, \infty)$ instead, $f_1(x) = x^2$ identifies θ_0 , and the method yields $\hat{\theta}_n = (\frac{1}{n} \sum_{i \leq n} X_i^2)^{1/2}$.

Part V. The method of moments

The method

Method of moments. We choose $f_1, \dots, f_p : \mathbb{R}^d \rightarrow \mathbb{R}$ and we look for a parameter $\hat{\theta}$ at which the theoretical moments equal the empirical ones:

$$\begin{cases} \frac{1}{n} \sum_{i \leq n} f_1(X_i) = \mathbb{E}_{\hat{\theta}}[f_1(X_1)] \\ \vdots \\ \frac{1}{n} \sum_{i \leq n} f_p(X_i) = \mathbb{E}_{\hat{\theta}}[f_p(X_1)] . \end{cases}$$

Compactly. With the moment map e of the identifiability slide and $e_n = \left(\frac{1}{n} \sum_{i \leq n} f_1(X_i), \dots, \frac{1}{n} \sum_{i \leq n} f_p(X_i) \right)^\top$, the system takes the form $e(\hat{\theta}) = e_n$, so that $\hat{\theta} = e^{-1}(e_n)$ whenever the inverse is available.

Intuition. The law of large numbers yields $e_n \xrightarrow{\mathbb{P}} e(\theta_0)$, and since e^{-1} is continuous near $e(\theta_0)$, the solution should be close to θ_0 . The analysis below makes this precise and adds the rate.

Exercise, the gaussian model

Exercise. Let $\Theta = \mathbb{R} \times (0, \infty)$, $\theta = (m, \sigma^2)$ and $\mathcal{L}_\theta = \mathcal{N}(m, \sigma^2)$. With $f_1(x) = x$ and $f_2(x) = x^2$, find the estimator.

Solution. We have $\mathbb{E}_\theta[f_1(X_1)] = m$ and $\mathbb{E}_\theta[f_2(X_1)] = m^2 + \sigma^2$, so that the system is

$$\begin{aligned}\frac{1}{n} \sum_{i \leq n} X_i &= \hat{m}_n, \\ \frac{1}{n} \sum_{i \leq n} X_i^2 &= \hat{m}_n^2 + \hat{\sigma}_n^2.\end{aligned}$$

It has the unique solution $\hat{m}_n = \bar{X}_n$ and $\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i \leq n} X_i^2 - \bar{X}_n^2$, that is,

$$\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i \leq n} (X_i - \bar{X}_n)^2,$$

i.e. the empirical mean and the empirical variance. Their joint asymptotic normality follows from Part II, and the theorem below recovers it from the general statement.

Asymptotic normality

Theorem. With e as above, assume

1. θ_0 is in the interior of Θ and there is $\epsilon > 0$ with $B(\theta_0, \epsilon) \subset \Theta$,
2. e is continuously differentiable on $B(\theta_0, \epsilon)$ with $Je(\theta_0)$ invertible,
3. $\mathbb{E}_X[|f_j(X_1)|^2] < \infty$ for $j = 1, \dots, p$.

Then we can define a random vector $\hat{\theta}_n$ that solves the moment system with probability going to 1, and

$$\sqrt{n} \left(\hat{\theta}_n - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top} \right) ,$$

where Σ_f is the covariance matrix of the random vector $(f_1(X_1), \dots, f_p(X_1))$.

Proof, local inversion

Since e is continuously differentiable near θ_0 with $Je(\theta_0)$ invertible, the inverse function theorem yields open sets $U \ni \theta_0$ and $V \ni e(\theta_0)$ such that $e : U \rightarrow V$ is a C^1 diffeomorphism, and, for every $v = e(u) \in V$,

$$(Je^{-1})(v) = (Je(u))^{-1} .$$

In particular, at the point of interest,

$$(Je^{-1})(e(\theta_0)) = (Je(\theta_0))^{-1} .$$

We only use hypotheses 1 and 2 here, and they allow us to apply the delta method to $\phi = e^{-1}$ at $e(\theta_0)$.

Proof, existence with probability going to one

By hypothesis 3, each $f_j(X_i)$ is square integrable, hence integrable on a probability space, so that the law of large numbers applies coordinatewise:

$$e_n \xrightarrow{\mathbb{P}} e(\theta_0) .$$

Since V is open and contains $e(\theta_0)$, this yields $\mathbb{P}(e_n \in V) \rightarrow 1$, and we define

$$\hat{\theta}_n = \begin{cases} e^{-1}(e_n) & \text{if } e_n \in V \\ \text{an arbitrary value} & \text{if } e_n \notin V . \end{cases}$$

On the event $\{e_n \in V\}$, we have $e(\hat{\theta}_n) = e_n$, so that $\hat{\theta}_n$ solves the moment system, and this event has probability going to 1. The estimator is to be understood in this sense, since the system may have no solution for small n , which does not matter asymptotically.

Proof, removing the arbitrary value

We define a companion sequence that always lies in V ,

$$\tilde{e}_n = \begin{cases} e_n & \text{if } e_n \in V \\ e(\theta_0) & \text{if } e_n \notin V, \end{cases}$$

so that $e^{-1}(\tilde{e}_n)$ is always defined, and equals $\hat{\theta}_n$ on $\{e_n \in V\}$.

Since $\theta_0 = e^{-1}(e(\theta_0))$, the two sequences $\sqrt{n}(\hat{\theta}_n - \theta_0)$ and $\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0)))$ coincide on $\{e_n \in V\}$, hence for every $\epsilon > 0$

$$\mathbb{P} \left(\left\| \sqrt{n}(\hat{\theta}_n - \theta_0) - \sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0))) \right\| \geq \epsilon \right) \leq \mathbb{P}(e_n \notin V),$$

which goes to 0.

By asymptotic equivalence, it is thus enough to find the limit law of $\sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0)))$.

Proof, the delta method

Since the vectors $(f_1(X_i), \dots, f_p(X_i)), i \leq n$, are i.i.d., square integrable by hypothesis 3, with mean $e(\theta_0)$ and covariance Σ_f , the central limit theorem yields

$$\sqrt{n}(e_n - e(\theta_0)) \xrightarrow{d} \mathcal{N}(0, \Sigma_f) ,$$

and since $\tilde{e}_n$ differs from e_n only on an event of vanishing probability, asymptotic equivalence yields the same limit for $\sqrt{n}(\tilde{e}_n - e(\theta_0))$.

We now apply the delta method to $\phi = e^{-1}$, differentiable at $e(\theta_0)$, with $r_n = \sqrt{n}$, and we obtain

$$\begin{aligned} \sqrt{n}(e^{-1}(\tilde{e}_n) - e^{-1}(e(\theta_0))) &\xrightarrow{d} (Je(\theta_0))^{-1} \mathcal{N}(0, \Sigma_f) && \text{by local inversion} \\ &= \mathcal{N}(0, (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top}) . \end{aligned}$$

Together with the removal of the arbitrary value, this concludes the proof.

Exercise, the gaussian model, continued

Exercise. In the gaussian model with $f_1(x) = x$, $f_2(x) = x^2$ and $\theta_0 = (m_0, \sigma_0^2)$, compute the asymptotic covariance of $\sqrt{n}((\hat{m}_n, \hat{\sigma}_n^2) - (m_0, \sigma_0^2))$ given by the theorem.

Solution. Since $e(m, \sigma^2) = (m, m^2 + \sigma^2)^\top$, the inverse of $Je(\theta_0)$ has rows $(1, 0)$ and $(-2m_0, 1)$. Using $\mathbb{E}_X[X_1^3] = m_0^3 + 3m_0\sigma_0^2$ and $\mathbb{E}_X[X_1^4] = m_0^4 + 6m_0^2\sigma_0^2 + 3\sigma_0^4$, we obtain

$$\Sigma_f = \begin{pmatrix} \sigma_0^2 & 2m_0\sigma_0^2 \\ 2m_0\sigma_0^2 & 4m_0^2\sigma_0^2 + 2\sigma_0^4 \end{pmatrix}.$$

Hence, we have

$$\begin{aligned} (Je(\theta_0))^{-1} \Sigma_f (Je(\theta_0))^{-\top} &= \begin{pmatrix} \sigma_0^2 & 2m_0\sigma_0^2 \\ 0 & 2\sigma_0^4 \end{pmatrix} \begin{pmatrix} 1 & -2m_0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \sigma_0^2 & 0 \\ 0 & 2\sigma_0^4 \end{pmatrix}. \end{aligned}$$

We recover the limit $\mathcal{N}(0, 2\sigma_0^4)$ of Part II for the empirical variance, and the two estimators are asymptotically independent, since the limit is a gaussian vector with diagonal covariance.

Reading the asymptotic variance

Let us take $p = 1$, so that, with $e(\theta) = \mathbb{E}_\theta[f_1(X_1)]$, the asymptotic variance is the number

$$\frac{\text{Var}_X(f_1(X_1))}{(e'(\theta_0))^2}.$$

The numerator is the noise. The smaller the fluctuation of $f_1(X_1)$, the more precisely the empirical moment estimates the theoretical one.

The denominator is the signal. The larger $|e'(\theta_0)|$, the more the theoretical moment moves when the parameter moves, and thus the better f_1 discriminates between θ_0 and its neighbours.

A good moment function thus has an expectation that varies fast with θ and a small variance under $\mathcal{L}_{\theta_0}$. However, the method does not prescribe f_1 , and different choices lead to different variances, none of them optimal in general.

Exercise, two moment functions

Exercise. Let $\Theta = (0, \infty)$ and $\mathcal{L}_\theta$ the exponential law of parameter θ , for which $\mathbb{E}_\theta[X_1^k] = k!/\theta^k$. Compute the asymptotic variance of the method of moments with $f_1(x) = x$, then with $f_1(x) = x^2$, and compare.

With $f_1(x) = x$, we have $e(\theta) = 1/\theta$, so that $e'(\theta_0)^2 = 1/\theta_0^4$, and $\text{Var}_X(X_1) = 1/\theta_0^2$. The asymptotic variance is thus θ_0^2 , as in the exercise on the inverse of a mean, since $\hat{\theta}_n = 1/\bar{X}_n$.

With $f_1(x) = x^2$, we have $e(\theta) = 2/\theta^2$, so that $e'(\theta_0)^2 = 16/\theta_0^6$, and $\text{Var}_X(X_1^2) = (24 - 4)/\theta_0^4$. Hence, the asymptotic variance is

$$\frac{20/\theta_0^4}{16/\theta_0^6} = \frac{5}{4}\theta_0^2.$$

Comparison. The first moment is better for every θ_0 , by a factor $5/4$. Indeed, moving from x to x^2 multiplies the standard deviation of the noise by $\sqrt{20}/\theta_0$, but the signal $|e'(\theta_0)|$ only by $4/\theta_0$.

The limits of the method

Example. Let $\Theta = (0, \infty)$, $\mathcal{L}_\theta = \text{Unif}([0, \theta])$ and $f_1(x) = x$. Then $e(\theta) = \theta/2$, so that $e'(\theta_0) = 1/2$ is invertible, $\Sigma_f = \text{Var}_X(X) = \theta_0^2/12$, and the theorem applies to $\hat{\theta}_n = 2\bar{X}_n$:

$$\sqrt{n} \left(\hat{\theta}_n - \theta_0 \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\theta_0^2}{3} \right) .$$

Comparison with the maximum of the sample. For $M_n = \max_{i \leq n} X_i$, a direct computation yields $\mathbb{P}(n(\theta_0 - M_n) > t) = (1 - t/(n\theta_0))^n \rightarrow e^{-t/\theta_0}$, so that $n(\theta_0 - M_n)$ converges in law to an exponential. Hence, the rate is $1/n$ rather than $1/\sqrt{n}$.

Interpretation. The method of moments is consistent and asymptotically normal here, but badly suboptimal, since the information about θ_0 lies in the largest observation rather than in an average. No choice of moment functions repairs this.